Differential Equations

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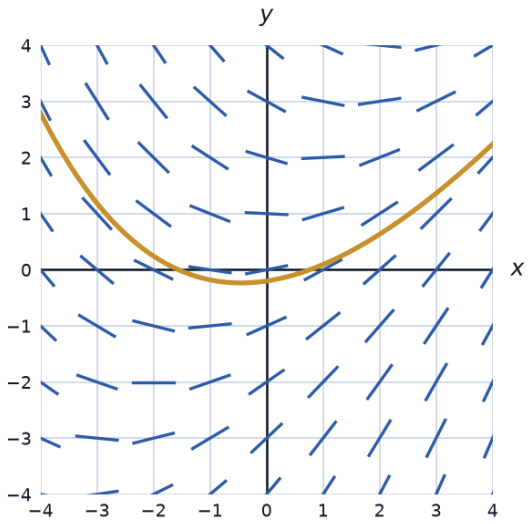
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Solving by separation of variables

Solve $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$.

2 [Foundational] • Target: Solving by separation of variables

Solve $\frac{dy}{dx} = 3x^2y$, given $y = 2$ when $x = 0$.



3 [Foundational] • Target: Solving by separation of variables

Solve $\frac{dy}{dx} = \frac{2x}{y^2}$, given $y = 2$ when $x = 0$.

4 [Foundational] • Target: Solving by separation of variables

Solve $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$, given $y = 1$ when $x = 1$.

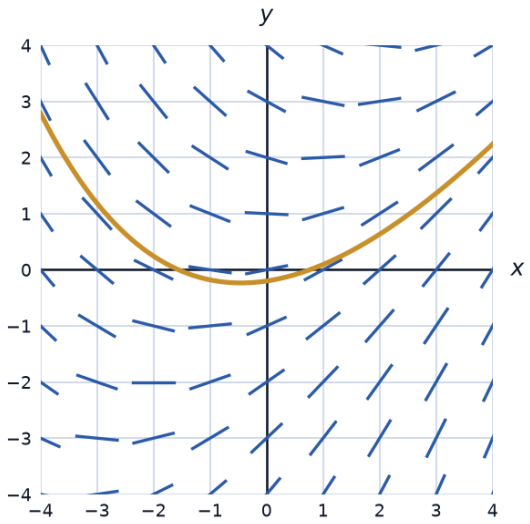
5 [Foundational] • Target: Forming differential equations from a description

The rate of decrease of a quantity Q is proportional to Q itself. Write a differential equation for $\frac{dQ}{dt}$, using $k > 0$ as the constant of proportionality.

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6 [Foundational] • Target: Forming differential equations from a description

A tank drains so that the rate of decrease of its volume V is proportional to $\sqrt{V}$. Write a differential equation for $\frac{dV}{dt}$, using $k > 0$.



7 [Foundational] • Target: Forming differential equations from a description

A chemical reaction's concentration C decreases at a rate proportional to the square of the concentration. Write a differential equation for $\frac{dC}{dt}$, using $k > 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving modeling equations with initial conditions

A population P satisfies $\frac{dP}{dt} = 0.08P$. If $P = 500$ when $t = 0$, find P when $t = 10$, to the nearest whole number.

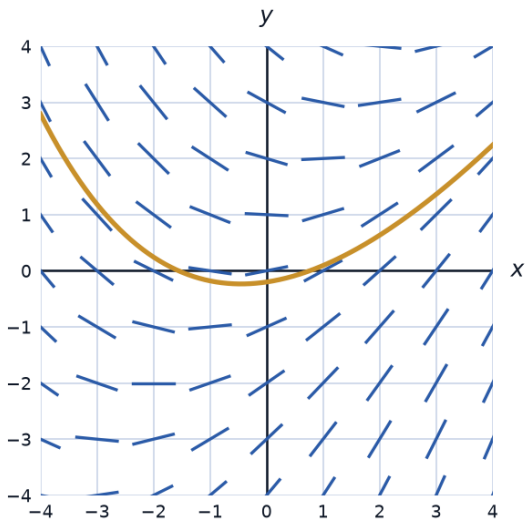
9 [Intermediate] • Target: Solving modeling equations with initial conditions

Solve $\frac{dV}{dt} = -k\sqrt{V}$ (from the tank problem above), given $V(0) = V_0$.

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10 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} + 3y = 12$ using an integrating factor, given $y = 2$ when $x = 0$.



11 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} - 2y = e^{3x}$ using an integrating factor, given $y = 1$ when $x = 0$.

12 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} + y = 5$ using an integrating factor, given $y = 0$ when $x = 0$.

13 [Intermediate] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} + 4y = 8$ using an integrating factor, given $y = 3$ when $x = 0$.

14 [Intermediate] • Target: Further integrating factor examples

Solve $\frac{dy}{dx} + \frac{y}{x} = x^2$ for $x > 0$ using an integrating factor, given $y = 1$ when $x = 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

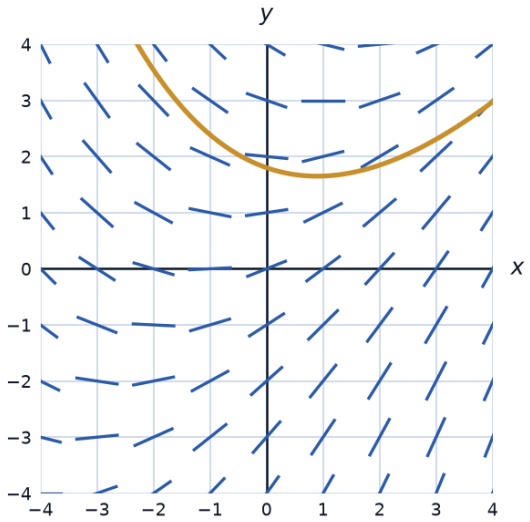
15 [Challenging] • Target: Further integrating factor examples • Cross-Lesson: 12.01 Poisson Distribution

Solve $\frac{dy}{dx} - \frac{2y}{x} = x^3$ for $x > 0$ using an integrating factor, given $y = 3$ when $x = 1$.

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16 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 13.01 Continuous Random Variables

Verify that $y = 3e^{2x}$ satisfies $\frac{dy}{dx} - 2y = 0$.



17 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 14.01 Sampling & Estimation

Verify that $y = x^2 + Ce^{-x}$ is a general solution of $\frac{dy}{dx} + y = x^2 + 2x$.

18 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 15.01 Hypothesis Testing

Verify that $y = \frac{1}{2}\sin(2x)$ satisfies $\frac{d^2y}{dx^2} + 4y = 0$.

19 [Challenging] • Target: General and particular solutions • Cross-Lesson: 2.01 Exponentials & Logarithms

Find the general solution of $\frac{dy}{dx} = 4x^3 - 6x$, and explain what determines a particular solution from this family.

20 [Challenging] • Target: General and particular solutions • Cross-Lesson: 3.01 Advanced Trigonometry

A curve satisfies $\frac{dy}{dx} = 6x^2 - 4$ and passes through $(1, -1)$. Find the equation of this particular solution.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Separation of Variables

Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$ with initial condition $y(0) = 4$.

a. $y^2 = 2x^2 + 16$

b. $y^2 = x^2 + 16$

c. $y = 2x + 4$

d. $y^2 = 2x^2 + 4$

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22 [Foundational] • Target: Separation of Variables

Find the general solution of $\frac{dy}{dx} = xy$.

a. $y = Ae^{x^2/2}$

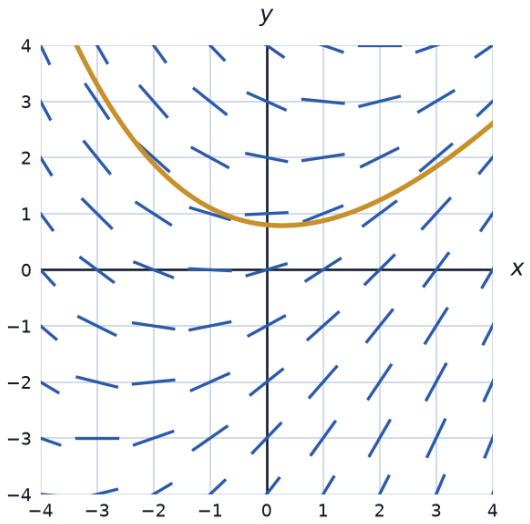
b. $y = Ae^{x^2}$

c. $y = \frac{1}{2}x^2 + C$

d. $y = Ae^{x}$

23 [Foundational] • Target: Separation of Variables

Solve the differential equation $\frac{dy}{dx} = 1 + y^2$.



a. $y = \tan(x + C)$

b. $y = \ln(1 + x) + C$

c. $y = e^x - 1$

d. $y = \arctan(x) + C$

24 [Foundational] • Target: First-Order Equations Requiring Partial Fractions

Find the general solution of $\frac{dy}{dx} = y(1 - y)$.

a. $y = \frac{Ae^x}{1 + Ae^x}$

b. $y = Ae^x - A$

c. $y = \ln|1 - y| + x$

d. $y = \frac{1}{1 + Ae^{-x}}$

25 [Foundational] • Target: First-Order Equations Requiring Partial Fractions

Solve $\frac{dy}{dt} = k(M - y)$ with $y(0) = 0$ for constant $k, M > 0$.

a. $y = M(1 - e^{-kt})$

b. $y = Me^{-kt}$

c. $y = M - e^{kt}$

d. $y = M(1 + e^{-kt})$

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26 [Foundational] • Target: First-Order Equations Requiring Partial Fractions

A chemical reaction produces a substance according to $\frac{dx}{dt} = 2(4 - x)^2$ with $x(0) = 0$. Find $x(t)$.

a. $x(t) = 4 - \frac{4}{8t+1}$

b. $x(t) = 4(1 - e^{-2t})$

c. $x(t) = \frac{8t}{2t+1}$

d. $x(t) = 4 - \frac{1}{2t}$

27 [Foundational] • Target: Rate of Change Modeling Problems

The rate of decay of a substance is proportional to its remaining mass m. Write down the differential equation governing $m(t)$ for constant $k > 0$.

a. $\frac{dm}{dt} = -km$

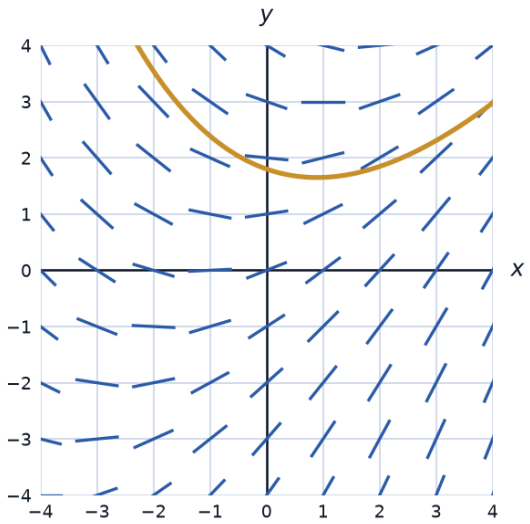
b. $\frac{dm}{dt} = km$

c. $\frac{dm}{dt} = -k$

d. $\frac{d^2m}{dt^2} = -km$

28 [Foundational] • Target: Rate of Change Modeling Problems

Water leaks from a cylindrical tank of cross-sectional area A through a hole at the bottom according to Torricelli's Law: $\frac{dh}{dt} = -c\sqrt{h}$. If initial height is H, how long does it take to empty completely?



a. $T = \frac{2\sqrt{H}}{c}$

b. $T = \frac{\sqrt{H}}{c}$

c. $T = \frac{H}{c}$

d. $T = \frac{H^2}{2c}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Rate of Change Modeling Problems

A population P grows according to the logistic model $\frac{dP}{dt} = 0.04P\left(1 - \frac{P}{2000}\right)$. What is the carrying capacity of the population?

a. 2000

b. 0.04

c. 1000

d. 50000

30 [Intermediate] • Target: Particular Solutions & Long-Term Behavior

For the logistic model in the previous question, at what population size P is the growth rate $\frac{dP}{dt}$ maximized?

a. $P = 1000$

b. $P = 2000$

c. $P = 500$

d. $P = 1500$

31 [Intermediate] • Target: Particular Solutions & Long-Term Behavior

Solve the differential equation $x\frac{dy}{dx} = y + 1$ with $y(1) = 3$.

a. $y = 4x - 1$

b. $y = 3x$

c. $y = 4x^2 - 1$

d. $y = 2x + 1$

32 [Intermediate] • Target: Particular Solutions & Long-Term Behavior

A body cools according to $\frac{dT}{dt} = -0.1(T - 20)$. What does $T(t)$ approach as $t \to \infty$?

a. $20\,^\circ\text{C}$

b. $0\,^\circ\text{C}$

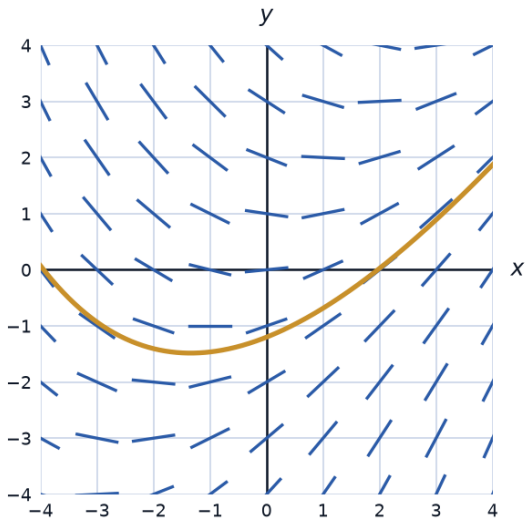
c. ∞

d. $-20\,^\circ\text{C}$

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33 **[Intermediate]** • **Target: Particular Solutions & Long-Term Behavior**

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 23$ and $T_0 = 63$. Find T at the time t when $e^{-kt} = 1/4$.



a. 33

b. 63

c. 34

d. 13

34 **[Intermediate]** • **Target: Solving by separation of variables**

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 25$ and $T_0 = 41$. Find T at the time t when $e^{-kt} = 1/4$.

a. 41

b. 21

c. 29

d. 30

35 **[Intermediate]** • **Target: Forming differential equations from a description**

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 25$ and $T_0 = 37$. Find T at the time t when $e^{-kt} = 1/4$.

a. 37

b. 28

c. 29

d. 22

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36 [Intermediate] • Target: Solving modeling equations with initial conditions

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 17$ and $T_0 = 27$. Find T at the time t when $e^{-kt} = 1/2$.

a. 27

b. 22

c. 12

d. 23

37 [Intermediate] • Target: First-order linear differential equations using an integrating factor

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 19$ and $T_0 = 31$. Find T at the time t when $e^{-kt} = 1/3$.

a. 23

b. 24

c. 15

d. 31

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Further integrating factor examples • Cross-Lesson: 15.01 Hypothesis Testing

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 16$ and $T_0 = 28$. Find T at the time t when $e^{-kt} = 1/2$.

a. 23

b. 22

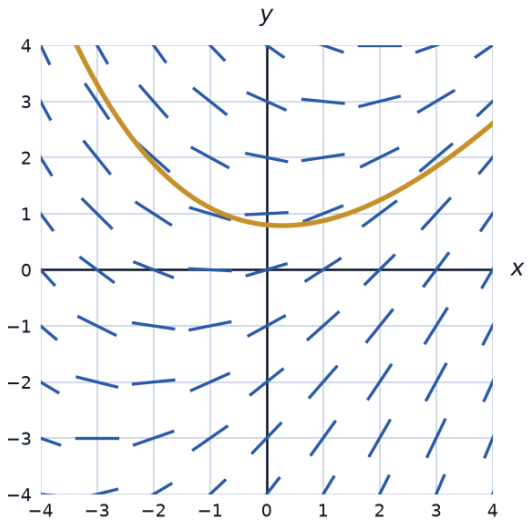
c. 28

d. 10

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39 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 2.01 Exponentials & Logarithms

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 18$ and $T_0 = 36$. Find T at the time t when $e^{-kt} = 1/3$.



a. 25

b. 12

c. 36

d. 24

40 [Challenging] • Target: General and particular solutions • Cross-Lesson: 3.01 Advanced Trigonometry

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 19$ and $T_0 = 35$. Find T at the time t when $e^{-kt} = 1/4$.

a. 35

b. 24

c. 15

d. 23

41 [Challenging] • Target: Applications: Newton's law of cooling • Cross-Lesson: 4.01 Differentiation

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 13$ and $T_0 = 37$. Find T at the time t when $e^{-kt} = 1/6$.

a. 9

b. 18

c. 17

d. 37

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42 [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 5.01 Integration

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 24$ and $T_0 = 48$. Find T at the time t when $e^{-kt} = 1/3$.

a. 16

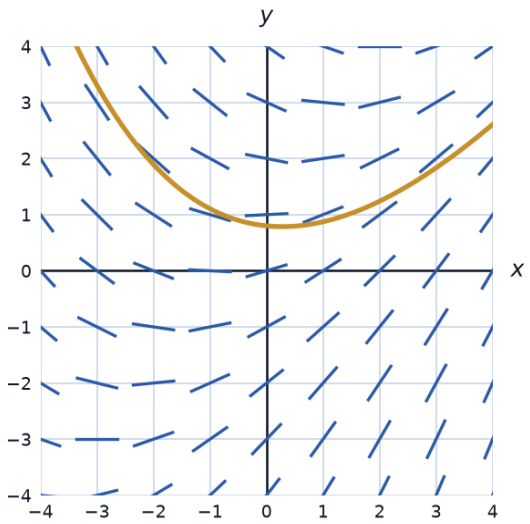
b. 32

c. 33

d. 48

43 [Challenging] • Target: Separation of Variables • Cross-Lesson: 6.01 Sequences & Series

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 24$ and $T_0 = 36$. Find T at the time t when $e^{-kt} = 1/6$.



a. 22

b. 36

c. 27

d. 26

44 [Challenging] • Target: First-Order Equations Requiring Partial Fractions • Cross-Lesson: 7.01 Vectors

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 23$ and $T_0 = 35$. Find T at the time t when $e^{-kt} = 1/3$.

a. 27

b. 19

c. 35

d. 28

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45 [Challenging] • Target: Rate of Change Modeling Problems • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 21$ and $T_0 = 45$. Find T at the time t when $e^{-kt} = 1/3$.

a. 13

b. 30

c. 29

d. 45

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